

Introduction to Scientific Computing

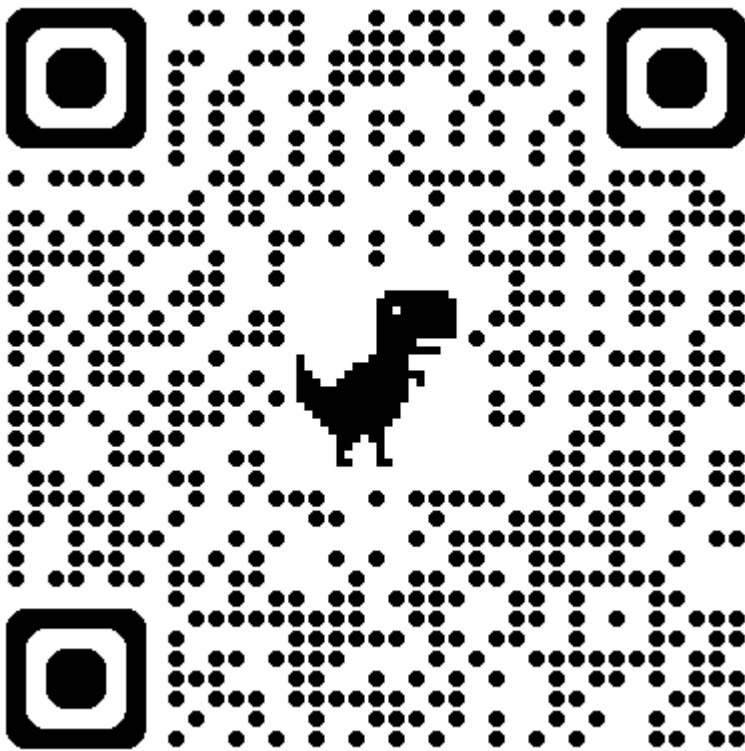
ENGR1050 – Lecture 20

Fall 2025

Nat Trask

University of Pennsylvania

Introduction to AI-assisted programming



Exam post-mortem

Basic Concepts

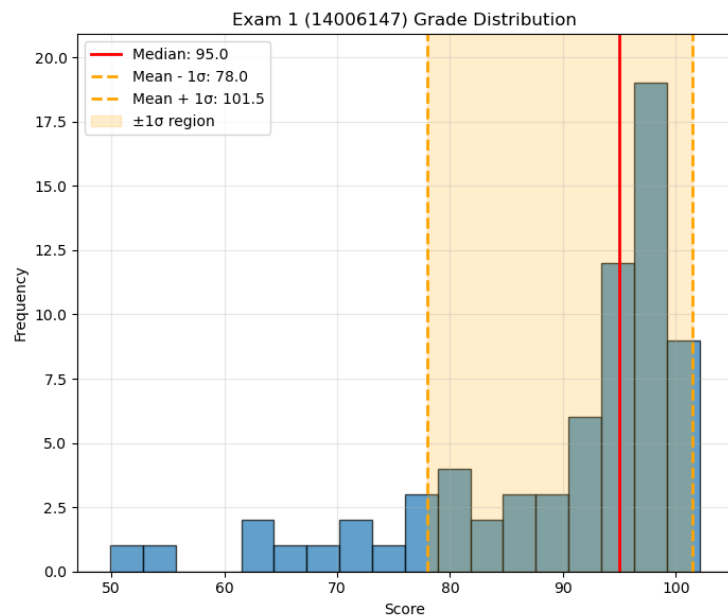


Intro to vibe coding

Alternatives to lists

In-class exercise

Practice what we've seen



Exam 1

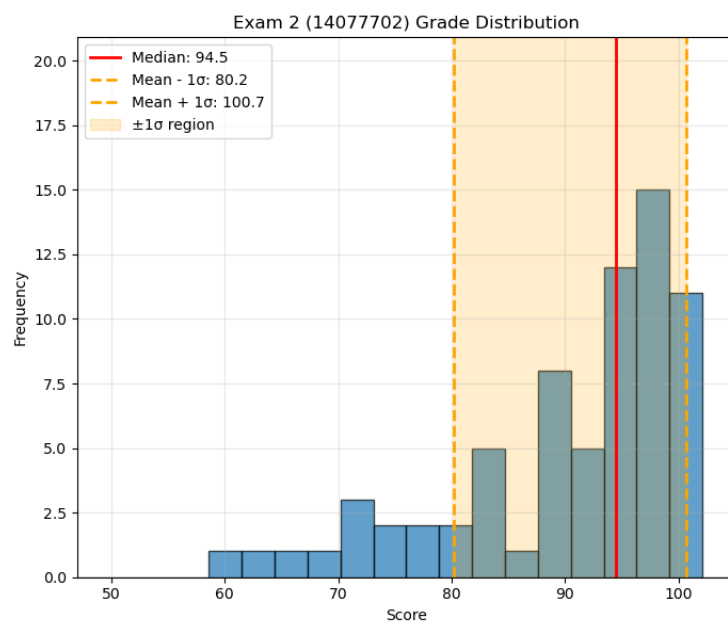
Mean: 89.74

Median: 95.00

Std Dev: 11.72

Min: 52.00

Max: 100.00



Exam 2

Mean: 90.44

Median: 94.50

Std Dev: 10.21

Min: 61.00

Max: 100.00

Problem 1: Definition of an ODE (25 points) Prof. Trask gives you the following ODE

$$\ddot{y} = -y$$

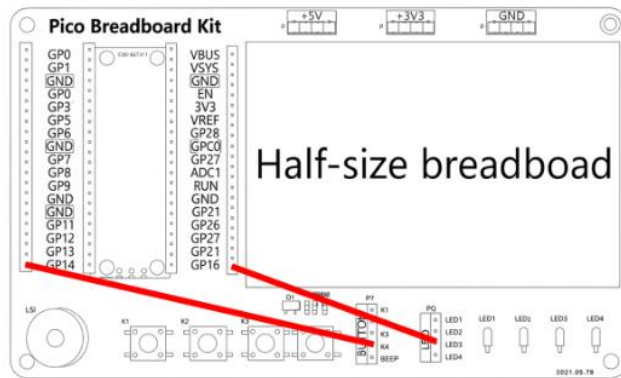
and claims that

$$y(t) = \cos t$$

is a solution.

- (a) This ODE is missing initial conditions. Provide a set of initial conditions at time $t = 0$ that would make Prof. Trask's claim true.
- (b) Verify that $y(t) = \cos t$ is indeed a solution.
- (c) Write the second-order ODE as a system of two first-order ODEs.

Problem 2: Pico buttons and LEDs (25 points) Somebody provides you with a Pico wired according to the following diagram:



Complete the TODO items in the following code snippet. The code should use functions from the provided classes to toggle the LED each time the button is pressed.

```
mybutton = ButtonClass(???) # TODO - properly instantiate ButtonClass
myled = LEDClass(???)      # TODO - properly instantiate LEDClass
while True:
    # Detect if a button was pressed
    if mybutton.catchPress():
        # If button was pressed, toggle the LED
        # TODO - add a single line of code here
```

```
class LEDClass:
    def __init__(self, pin_number, initial_state=False):
        self.led = Pin(pin_number, Pin.OUT)
        self.state = initial_state
        if self.state:
            self.led.on()
        else:
            self.led.off()

    def toggle(self):
        if self.state:
            self.led.off()
            self.state = False
        else:
            self.led.on()
            self.state = True
```

Problem 3: Solving with ODE_solver class. (25 points) You are provided the following numerical solver inheriting from the ODE_solver class.

```
class childclassODE(ODE_solver):
    def __init__(self, t0, tf, y0, N, newparameter):
        super().__init__(t0, tf, y0, N)
        self.newparameter = float(newparameter)
    def f(self, t, y):
        f = np.zeros(self.dim)
        f[0] = 1 - self.newparameter * y[0]**2
        return f
```

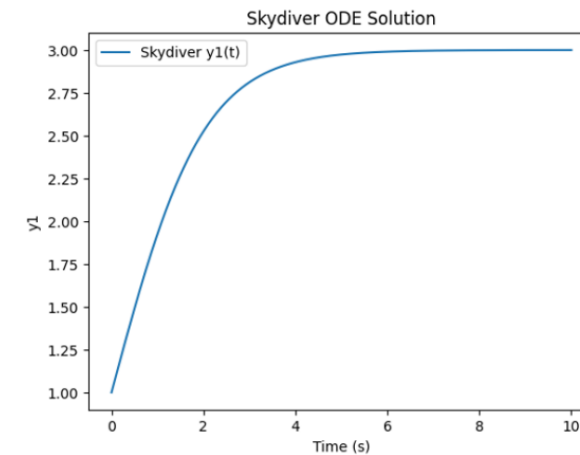
- (a) (10 points) What is the ODE being solved by this class? *Hint:* You can write it in the form $\dot{y} = f(t, y)$ by identifying $f(t, y)$ from the code.

- (b) (10 points) Derive a formula for the steady-state solution (i.e., when $\dot{y} = 0$) in terms of **newparameter**.

- (c) (10 points) Somebody initializes the class as follows:

```
solver = childclassODE(0, 10, [0], 1000, 1)
solution = solver.solve()
```

And produces the following plot of y versus time:



Is this solution reasonable? Explain why or why not.

Problem 4: Implementing explicit Euler. (25 points) The following is the definition of the explicit Euler method.

Definition

For the IVP $y'(t) = f(t, y)$ with $y(t_0) = y_0$ and step size $h > 0$, starting from (t_0, y_0) , the Explicit Euler scheme updates via

$$t_{n+1} = t_n + h, \quad y_{n+1} = y_n + h f(t_n, y_n).$$

Write a for loop that implements the explicit Euler method to solve the following ODE:

$$\dot{y} = -3y$$

Use the initial condition $y(0) = 5$, a time step size of $h = 0.1$, and compute the solution from $t = 0$ to $t = 2$. You can store the solution in either a list or a Numpy array.

Intro to AI for programming unit

Writing code with AI

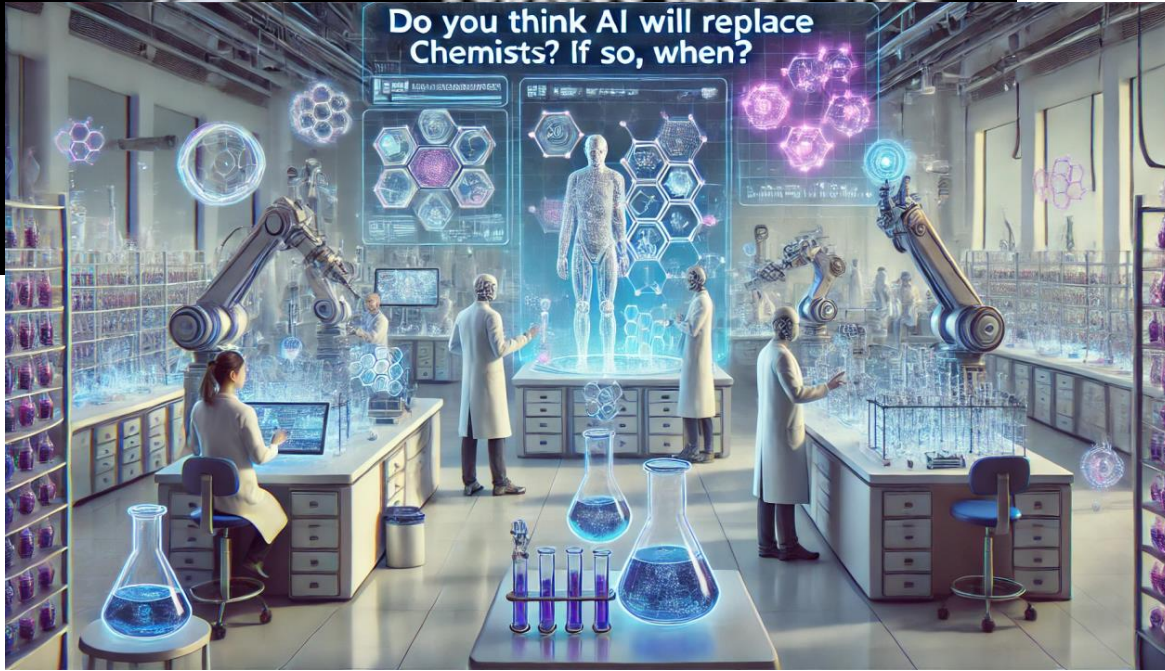
Using AI to learn new Python libraries

Running basic machine learning in Pytorch

AI Learning Lesson

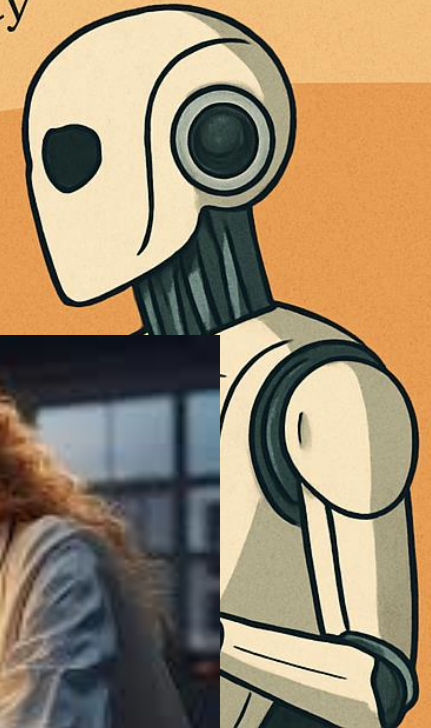
“AI won't replace humans — but humans who know how to use AI will replace those who

Do you think AI will replace Chemists? If so, when?



AI will not replace humans, but those who use AI will replace those who don't.

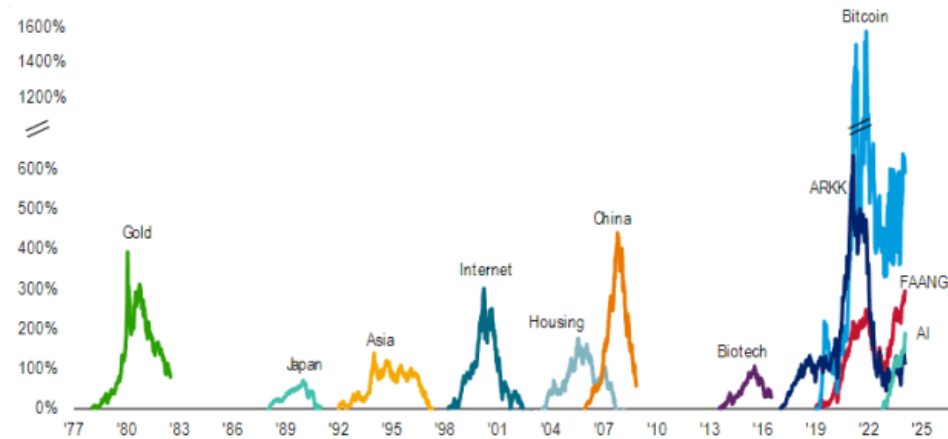
— Ginni Rometty



Why AI Won't Replace Humans:
A SCIENTIFIC PERSPECTIVE
NADER AL MALLAH

Chart 2: Tech Bubbles popped by Rates, Penetration and/or Regulation

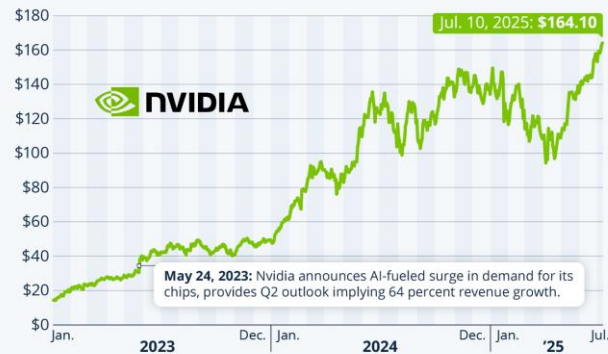
History of asset bubbles



Source: BofA Global Investment Strategy, Bloomberg

AI Boost: Nvidia Surges 1,000% in 30 Months

Nvidia's stock price since January 3, 2023*



* Daily closing prices adjusted for stock splits
Source: Yahoo Finance

statista

There is massive uncertainty around AI

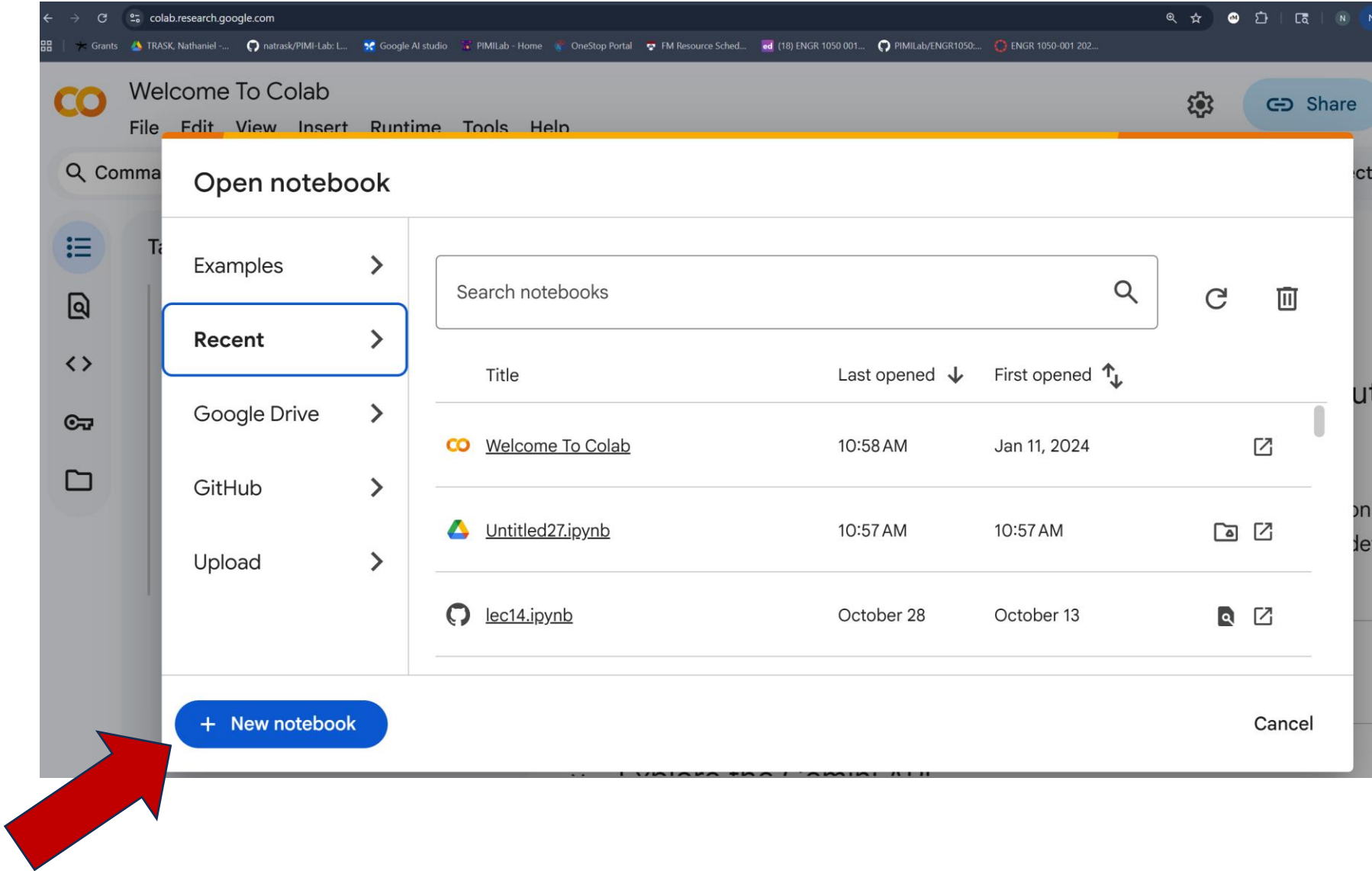
- Tools are here that weren't here 2 years ago
- Tools here today might not be as good 2 years in the future

What can we do to build strong fundamentals that future proof our skills as engineers?

Learning Goals
Literacy
Ethical Use
Technical Competency

Getting started with Gemini in Colab

Open Colab > Click New notebook



Click Settings icon > AI Assistance > Check inline completion + generative AI features

  Untitled28.ipynb ☆

File Edit View Insert Runtime Tools Help

Q Commands

+ Code ▾

+ Text

▶ Run all ▾

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[] | start coding or generate with AI.

Settings

Site >

☒ Show AI-powered inline completions

Editor >

☒ Consented to use generative AI features

AI Assistance >

☐ Hide generative AI features

Colab Pro >

Gemini can make mistakes so double-check responses and [use code with caution.](#)

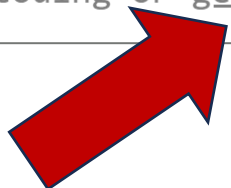
GitHub >

Two options: click generate with AI, or turn on Gemini chat dialog

[]



Start coding or generate with AI.

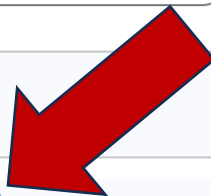


[How can I install Python libraries?](#)

[Load data from Google Drive](#)

[Show an example of training a](#)

What can I help you build?



ables

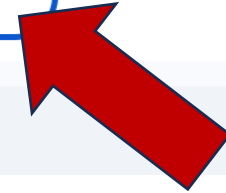
Terminal

A dialog will pop up that will write markdown+code for you

Show me how to install the Gemini API library

Explain responsible AI usage guidelines for Gemini

i am a beginner to python taking a intro to python for engineers class. generate a simple class that implements a explicit euler to solve a first order ode



✓ Explicit Euler Method

The explicit Euler method is one of the simplest numerical methods for approximating the solution of ordinary differential equations (ODEs) with a given initial value. It's a first-order method, meaning that the local error is proportional to the square of the step size, and the global error is proportional to the step size.

Given a first-order ODE of the form: $\frac{dy}{dt} = f(t, y)$ with an initial condition $y(t_0) = y_0$,

The explicit Euler method approximates the solution $y(t)$ at future time steps using the formula: $y_{i+1} = y_i + h * f(t_i, y_i)$

Where:

- y_i is the approximate value of the solution at time t_i .
- y_{i+1} is the approximate value of the solution at the next time step t_{i+1} .
- h (often denoted as dt) is the step size ($t_{i+1} - t_i$).
- $f(t_i, y_i)$ is the value of the derivative at t_i and y_i .

Todays task: Vibe-coding data collection

docs.google.com/forms/u/0/?tgif=d

Grants TRASK, Nathaniel - ... natrask/PIMI-Lab: L... Google AI studio PIMILab - Home OneStop Portal FM Resource Sched... (18) ENGR 1050 001... PIMILab/ENGR1050: ENGR 1050-001 202...

Forms Search

Start a new form

Blank form

Blank Quiz

Exit Ticket

Assessment

Worksheet

Template gallery

Recent forms

Owned by anyone

ENGR1050 Check in

ENGR1050 - End of unit 1 check in

ENGR1050 - Lecture 01

ENM5320

ENGR1050 Check in #2

ENGR1050 - End of unit 1 ...

ENGR1050 - Lecture 01

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Responses > View in Sheets



ENGR1050 Check in #2



Published

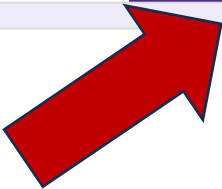


Questions

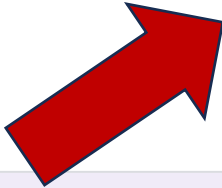
Responses **33**

Settings

33 responses



View in Sheets



Summary

Question

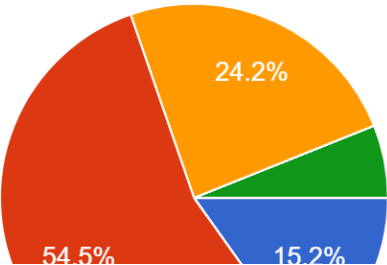
Individual

How difficult are you finding the course so far?



Copy chart

33 responses



- Challenging, and I don't feel like I'm getting the resources I need to succeed
- Challenging, but have the resources to push myself and learn the material
- Middle of the road for a UPenn course
- Easy, but I don't mind
- Too easy, there should be more material

Extracting data from this spreadsheet

 ENGR1050 Check in #2 (Responses) ☆ 📁 ☁

File Edit View Insert Format Data Tools Extensions Help Ask Gemini

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	A	B	C	D	E
	Form_Responses ▾				
1	Timestamp ▾	How difficult are you finding the course so far ▾	How is the pace of the course? ▾	Whether you like programming or not, are you ▾	How do you feel about more simulation ▾
2	10/13/2025 14:28:58	Challenging, but have the resources to push myse	A little fast but manageable	I'm happy with the class	Current plan is ok (50%/50%)
3	10/13/2025 14:29:32	Challenging, but have the resources to push myse	A little fast but manageable	Unhappy; this isn't what an intro to python for eng	Current plan is ok (50%/50%)
4	10/13/2025 14:29:37	Middle of the road for a UPenn course	A little fast but manageable	I'm happy with the class	Current plan is ok (50%/50%)
5	10/13/2025 15:17:10	Easy, but I don't mind	A little slow but you're ok with it	I'm happy with the class	More data science/less simulation
6	10/13/2025 15:28:08	Challenging, and I don't feel like I'm getting the res	A little fast but manageable	This is fine (dog on fire drinking coffee)	Current plan is ok (50%/50%)
7	10/13/2025 15:35:13	Middle of the road for a UPenn course	A little fast but manageable	I'm happy with the class	More data science/less simulation
8	10/13/2025 15:35:17	Challenging, but have the resources to push myse	Too fast!	This is fine (dog on fire drinking coffee)	Current plan is ok (50%/50%)
9	10/13/2025 15:36:44	Challenging, and I don't feel like I'm getting the res	Too fast!	Unhappy; this isn't what an intro to python for eng	More data science/less simulation
10	10/13/2025 15:37:43	Challenging, and I don't feel like I'm getting the res	Too fast!	Unhappy; this isn't what an intro to python for eng	

Next homework

